DAY - 13

SEAT NUMBER | M | 0 | 0 | 3 | 4 | 8 |

2012	III	16	1100	V – 668	(E)

COMPUTER SCIENCE

PAPER - II (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : (1) *All questions are compulsory.*
(2) *Figures to the right indicate full marks.*
(3) *Draw neat diagrams wherever necessary.*
(4) *Use of calculator is not allowed.*
(5) *Due credit will be given for the programs with appropriate comments/ remarks.*

(A) Select correct alternatives and rewrite the following :

(a) In 8085 Microprocessor, _____ pin is the only output terminal of interrupt control block. 1

(i) TRAP
(ii) INTR
(iii) RST 7.5
(iv) $\overline{INTA}$

(b) _____ is a characteristic feature of 8051 Micro-controller. 1

(i) Four 8 bit I/O Ports
(ii) Two 8 bit I/O Ports
(iii) 4kB RAM
(iv) Four External Interrupts

(c) The instruction JNZ of 8085 Microprocessor is _____ type of Instruction. 1

(i) Branching
(ii) Conditional Branching
(iii) Arithmetic
(iv) Data Transfer

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(d) The Installation Cost of _____ Cable is maximum. 1

(i) STP

(ii) UTP

(iii) Fiber Optic

(iv) Co-axial

(B) Answer **any two** of the following :

(a) Explain any three primary functions of the CPU of a Micro-computer. 3

(b) Explain with suitable examples the (i) Register Indirect and (ii) Implied Addressing Modes of 8085 Microprocessor. 3

(c) Explain in brief the six important characteristics of Transmission Media. 3

(A) Answer **any two** of the following :

(a) In case of a Microprocessor Architecture, explain the following terms in brief : 3

(i) Address Bus

(ii) Data Bus

(iii) Control Bus

(b) Explain the following instructions of 8085 Microprocessor with suitable examples : 3

(i) SBB M

(ii) CPI Data

(iii) XTHL

(c) Explain Fibre Optic Cable with a suitable figure. 3

(B) Answer **any one** of the following :

(a) Compare any four attributes of 486 DX with Pentium Processor. 4

(b) Explain use of extended register pairs BC and HL of 8085 Microprocessor as Pointers with the help of suitable examples. 4

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3. (A) Answer **any two** of the following :

(a) Describe in brief the function of the following pins of 8085 Microprocessor : 3

(i) $\overline{RD}$

(ii) $\overline{WR}$

(iii) ALE

(b) The accumulator of 8085 contains data $B7_H$. What will be its contents after execution of the following instructions independently ? 3

(i) ORI 58_H

(ii) CMA

(iii) ANI $E3_H$

(c) Compare at least three characteristics of UTP and STP Cables. 3

(B) Answer **any one** of the following :

(a) What are Vectored Interrupts ? What are Maskable and Non-maskable Interrupts ? State all Hardware Interrupts of 8085 Microprocessor with their priorities and branching or vector addresses. 4

(b) Give any eight important features of 8051 Microcontroller. 4

4. (A) Answer **any two** of the following :

(a) Explain the following instructions of 8085 Microprocessor with suitable example : 3

(i) DAA

(ii) DAD

(b) Explain Stack Operation in case of 8085 Microprocessor with the help of suitable instructions like PUSH and POP. 3

(c) Compare 8052 Microcontroller with 8051 Microcontroller. 3

(B) Answer **any one** of the following :

(a) Explain STAR Topology with necessary diagram. State its advantages. 4

(b) Explain the following devices used in Computer Networking : 4

(i) Modem

(ii) Repeater

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5. Answer **any two** of the following :

(a) An 8 bit number is stored in memory location 4400H. Write an assembly language program to count 'Zero' in the given number. Store count in memory location 4500H. 5

(b) A series of numbers are stored in memory locations from C001H to C008H. Write a program in assembly language to find smallest number among these numbers. Store smallest number in memory location C009H. 5

(c) Write an assembly language program to count number of odd data bytes occurring in a block starting from memory location A001H to [illegible] Store result at memory location B000H. 5

OR

5. Answer **any two** of the following :

(a) A Hex number is stored at location AB00H. Write an assembly language program to interchange its digit. The new number is to be stored at AB01H. Add original number with new number and store result at location ABCDH. 5

(b) Write an assembly language program to add two BCD numbers stored at locations AB00H and AB01H. Place BCD result in location AB02H and onwards starting with LSB. 5

(c) Write a program in assembly language to find 2's Complement of 8 bit number stored in memory location C000H. Store result at memory location C001H. 5

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